RGCS v3 Software and Hardware Architecture and Roadmap

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Abstract

The v3 platform layers a reusable typed mathematics library (`rscs_core`) under the deterministic crystal-application core (`rgcs_core`) and the desktop workbench, adds a synchronized excitation/measurement architecture with one master timebase, and defines an embedded platform contract (HG Embedded OS, **[ENG]**). The Windows portability debt of v2 is paid (POSIX bundle archive names, declared dependencies, a two-OS CI matrix), persistence layers exist for crystal records and Hydrogenuine memory, and the timing-hardware roadmap is quantified before any exotic hardware is proposed: a TCXO reference is mandatory, a DDS covers non-integer channels, a small CPLD is justified only as a safety function, and an FPGA is explicitly *not* justified. All unbuilt hardware is **[ENG]** until built and measured.

1 Layering

`rscs_core` (typed coordinates/operators, registry, embedding, firewall) is consumed by `rgcs_core` (geometry, resonance, anisotropy, optics, timing, drive, experiments, persistence, FEA export) which is consumed by `rgcs_desktop` (Qt panels as thin views over headless, tested services). The mathematical provenance graph (76 nodes, 76 edges across the three machine registries) is itself a generated artifact: every UI element that shows a value shows its classification.

2 Timing architecture

One authoritative master timebase (32.768 kHz TCXO $\div 8 \to$ 4096 Hz carrier) derives every channel; a channel without a *measured* latency calibration is phase-invalid (gate H-29). Exact-cycle closures of the source keys **[SRC]** are pure arithmetic **[DER]**:

Table 1: Exact-cycle closures of the source key frequencies **[SRC]** against the 4096 Hz carrier (`rgcs_core.timing.key_closures`); the closure arithmetic is **[DER]** and confers no truth value on the keys.

Key (Hz)	Window (ms)	Key cycles	Carrier cycles
1496	125	187	512
644	250	161	1024
587	1000	587	4096

Table 2: Modulation families: source rates vs. exact-cycle engineering variants (carrier subharmonics).

f (Hz)	Origin	Exact	Closure (ms)
20.00	Source claim (RG-13)	no	250.000
20.48	Derived exact variant	yes	48.828
21.00	Source claim (RG-13)	no	1000.000
40.96	Derived exact variant	yes	24.414

The three macro envelope modes are frozen v2 definitions with cycle-exact allocations; the source's ambiguous "shorter by half" statement maps to the two distinctly named modes `half_spacing` and `double_rate` (D7-002) — neither is ever renamed.

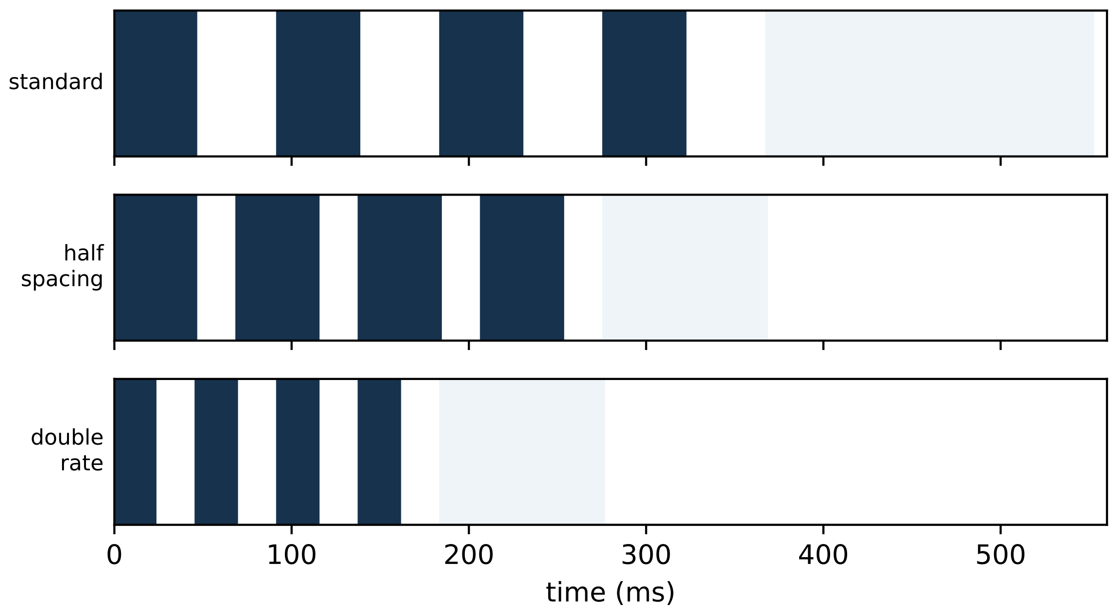

Figure 1: The three frozen macro envelope modes (dark = ON bursts, light = final pause), rendered by the same code the desktop preview and firmware acceptance tests use.

3 Phase at the interaction coordinate

The commanded phase is never the phase at the crystal. The platform accumulates six declared terms — cable, driver, inductive arctan$(\omega L/R)$, acoustic (oriented speed from the anisotropic model, never a guessed scalar), optical transit, and group delay — and reports each separately. The interaction coordinate itself is a named choice (geometric centre / predicted node / measured node), and coil current, voltage, field, and phase are measured *at the load*.

4 Safety envelope

Binding limits (D7-003): ≤ 30 V, ≤ 3 A peak, ≤ 5 mJ per pulse, laser class $\leq$ 3R at ≤ 5 mW with interlock, specimen temperature-rise stop, and *dummy-load-first*: no operating point reaches a specimen before it is cleared on an instrumented dummy load. There is no human-exposure or therapeutic protocol anywhere in the programme, and function-generator presets are signal-level recipes only.

5 Embedded platform (HG Embedded OS) [ENG]

An ESP32/CYD-class board running FreeRTOS with a strict layer contract: BSP (capabilities, no policy) $\rightarrow$ SDK (UI, storage, deterministic timing, peripherals, logging, lifecycle) $\rightarrow$ one build-time-selected app (RGCS controller, Karaoke migration as an SDK proof, self-test). Safety limits compile into firmware — SD-card configuration may tighten but never loosen them; arming outputs requires a closed hardware interlock loop, a passing boot self-test, and per-channel latency calibration. The application manifest schema rejects any app requesting an uncapped output class.

6 Timing-hardware roadmap (quantified)

Requirements first: jitter ≤ 100 ns rms, phase resolution $\leq 1°$ at 4096 Hz (678 ns), drift ≤ 2 ppm. Against these: ESP32 peripherals meet jitter and resolution but its crystal fails drift, so the **TCXO reference is mandatory**; an external **DDS** synthesizes the non-integer channels the master-clock model flags (1496/644/587/20/21 Hz); a **CPLD** is justified for the complementary coil pair with hardware dead-time — a safety function, not prestige; an **FPGA is not justified** at these rates and is deferred until a measurement requirement demonstrably exceeds the TCXO/DDS/CPLD chain.

7 Portability and reproducibility

v2's Windows defects are closed: bundle archive names are POSIX on every platform (the checksum-verification failure class is regression-guarded), and the "specimen-listing defect" proved to be an undeclared dependency — the honest lesson is that cross-platform reproducibility requires *declared* environments plus a CI matrix, which now runs Linux and Windows across two Python versions with schema-validation and generated-artifact freshness gates. One platform-documented exception remains: byte-equality of Linux-generated golden CSVs is not portable across platforms/library versions and is tolerance-checked instead (tests remain byte-exact on the Linux reference platform).

8 Implementation tranches

T1 (complete): fixes, CI, persistence, FEA contract, headless services, embedded contracts. T2: desktop panels over the tested services. T3: FEA import scripts and analysis views. T4: firmware (BSP, timing service, self-test). T5: timing hardware (TCXO/DDS/CPLD + interlocks; [ENG] until measured). T6: the H-29 latency-calibration campaign that unblocks phase claims. Dependencies: T1→T2→T3 and T1→T4→T5→T6.

9 Lessons learned (engineering)

- Independent QA must be allowed to overturn the project's own mathematics: the v2 coupling-map correction (anti-Hermitian $K = i2\pi g$) changed the physical interpretation and became a permanent regression test.
- Classification and provenance prevented source laundering: historical operating points parameterize protocols as [SRC] without ever becoming evidence.
- Frozen registries plus conservative-extension tests made a major version safe: v3 provably cannot silently change v2.
- Cross-platform reproducibility is a declared-environment problem before it is a numerics problem.
- The software and models advanced faster than the measurement programme — the strongest current output is a reproducible framework and falsification plan, not experimental confirmation, and the platform says so on every claim-bearing surface.

References